Intergenerational Housing Misallocation and Fertility

Compact Theory and Quantitative Lifecycle Model

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Abstract

Children require goods and housing space. The space requirement matters because family-sized housing is harder to access than small rental housing: large units are concentrated in ownership, ownership requires down payments, and tax rules can induce older owners to retain houses whose marginal value to them is below the value to constrained young families. This draft has two parts. Part I gives a compact two-period model that isolates the mechanism and delivers goods-equivalent wedges for young access constraints, old-owner lock-in, and fertility. The compact model also makes clear which aggregate birth effects come from completed fertility among entrants, which come from entry into the housing market, and which come from tenure switching. Part II embeds the same mechanism in a finite-life quantitative lifecycle model with income risk, tenure and house-size choice, down-payment constraints, transaction costs, bequests, child aging, and a single aggregate housing-services market.

1 Introduction

A child changes a household's housing problem. It is not only another set of goods to buy, or more time to spend at home. It is often a need for a different dwelling: another bedroom, a larger living space, or a unit where raising a child is feasible. In many high-cost cities this margin is hard to adjust. Large rental units are scarce, family-sized homes are concentrated in ownership, and ownership requires down payments and borrowing capacity at ages when many households have little wealth.

This paper studies whether those space and financial constraints affect final family size and the number of births generated by a cohort. That distinction matters. A housing shock can move births across years without changing completed fertility. It can also change local births by changing who enters a high-cost housing market, even if completed fertility among incumbent households moves little. The question here is when housing constraints change completed fertility, when they mainly change entry and local scale, and how both effects depend on who can access family-sized space over the lifecycle.

The mechanism has three parts. First, children raise the household's demand for space. If the rental market does not offer sufficiently large units, the shadow price of an additional room can exceed the posted rent. Second, the natural substitute is ownership, but buying a family-sized home requires liquidity and credit. A young household may value the larger unit and still be unable to purchase it. Third, older owners may remain in large homes even when their own marginal value of space is low, because transaction costs and tax rules subsidize staying put. Housing scarcity then shows up not only as a high price of space, but also as a mismatch between who occupies family-sized units and who values them for fertility.

I develop the argument in two steps. The first part of the paper gives a compact two-period model that isolates the mechanism. It expresses the rental space constraint, the ownership finance constraint, old-owner lock-in, and the fertility distortion as goods-equivalent wedges. The second part embeds the same economics in a finite-life quantitative model with income risk, tenure and house-size choice, down-payment constraints, transaction costs, bequests, child aging, and a single aggregate housing-services market. The model combines lifecycle fertility choice (Sommer, 2016; Doepke and Kindermann, 2019) with housing tenure choice, collateral constraints, and owner-occupied housing (Henderson and Ioannides, 1983; Sommer et al., 2013; Sommer and Sullivan, 2018). The compact housing block follows the one-market floorspace logic in Coven et al. (2025): all housing services clear in one market, while tenure, borrowing, and tax rules determine who can actually occupy family-sized units.

2 Related Literature

This paper contributes to three strands of literature.

First, it contributes to work on fertility decline, fertility choice, and the housing determinants of fertility. Since Becker (1960) and Becker and Lewis (1973), economics has treated children as choice variables that enter preferences and require resources. Quantitative versions of this approach put that choice into a lifecycle environment with income risk, savings, timing, and intrahousehold bargaining (Jones et al., 2010; Sommer, 2016; Doepke and Kindermann, 2019). Recent work on the U.S. fertility decline emphasizes that falling birth rates are hard to explain with a single postponement margin or one obvious period shock (Kearney et al., 2022). The housing literature shows that housing prices, housing wealth, and mortgage finance move fertility behavior: Dettling and Kearney (2014) find opposite house-price effects for owners and nonowners, Lovenheim and Mumford (2013) find that housing wealth raises homeowners' fertility, Cumming and Dettling (2024) show that mortgage-rate pass-through affects births through cash flows, and Hacamo (2021) finds that mortgage deregulation increased home purchases and childbearing for young households. The structural housing-fertility literature studies the same broad connection in equilibrium: Couillard (2025) models joint location, house-size, and fertility choice, and shows that the supply of large units matters for aggregate fertility.

The object in this paper is closer to completed fertility than to a current birth rate, so I separate timing evidence from final-family-size evidence within this same strand. Simon and Tamura (2009) use U.S. Census data from 1940–2000 and find that higher rents per room are associated with lower fertility, including in completed-fertility specifications for older households. Clark (2012) is an important caution: expensive U.S. housing markets delay first births by several years, but do not clearly reduce completed fertility. More recent evidence is more favorable to the family-size interpretation. Japaridze and Sayour (2024) link unaffordable homeownership to delayed entry into motherhood and lower completed fertility. Dettling and Kearney (2025) study the FHA and VA mortgage programs and find that low-down-payment, long-term mortgage access increased births during the U.S. baby boom; their Census evidence on children ever born is consistent with higher completed fertility for groups with access to the programs. van Doornik et al. (2025) exploit randomized Brazilian housing-credit lotteries and find larger lifetime fertility effects when housing access arrives earlier in the childbearing years. I use this literature to discipline the distinction between three objects: current births, completed fertility among entrants, and births generated by a cohort after entry into the housing market is allowed to respond.[1]

[1] The same issue appears in demography and life-course research, which documents that fertility, homeownership,

Second, the paper contributes to the literature on intergenerational housing allocation and lock-in. The closest paper is Coven et al. (2025). Their model studies how property taxes, capitalization, and financial constraints allocate housing between older homeowners and younger constrained households. This paper uses the same core allocation problem: young households may value family-sized space but face down-payment constraints, while older incumbents may retain housing that would be valuable to younger families. The departure is that I make children part of the household problem. Reallocating family-sized space is then not only a homeownership or welfare issue; it also changes the price of children, completed fertility among entrants, and local births through entry.

The lock-in mechanism is also related to work on mortgage lock-in and housing ladders. Fonseca and Liu (2023) show that locked-in mortgage rates reduce mobility and dampen labor reallocation. In a spatial housing-ladder model, Fonseca et al. (2026) show that lock-in reduces moves down the ladder and keeps missing downsizers in larger homes, with equilibrium effects on prices and rents. My lock-in force comes from transaction costs and the tax treatment of capital gains at death rather than from fixed-rate mortgages, but the allocation logic is the same: incumbent frictions keep housing with households whose current marginal value of space may be low, tightening access for younger households whose family choices are space-intensive.

Third, the paper contributes to structural work on housing, tenure choice, and spatial allocation. The housing block builds on models in which tenure, collateral constraints, and owner-occupied housing shape household behavior (Henderson and Ioannides, 1983; Sommer et al., 2013; Sommer and Sullivan, 2018). It also relates to spatial-equilibrium work in which housing supply and local prices allocate households across space (Rosen, 1979; Roback, 1982; Gyourko et al., 2013; Hsieh and Moretti, 2019). The compact model keeps one floorspace market and uses sufficient statistics for the allocation wedges. The quantitative model then adds lifecycle income risk, tenure and house-size choice, children aging out of the household, bequests, and housing supply. The aim is to connect the intergenerational allocation margins in the housing literature to the fertility objects emphasized in the first strand.

Part I

Compact Analytical Model

3 Environment

Agents. The economy is populated by overlapping generations of young and old households. A young household has type

$$i = (y_i, a_i),$$

where $y_i > 0$ is disposable income and $a_i \geq 0$ is entry wealth, inherited from the previous old generation. Conditional on entry, the young household chooses tenure, housing, completed fertility, and saving. Old households are indexed by j. An old incumbent is last period's young owner and enters old age with financial wealth, the house she bought, and an accrued taxable gain. She chooses

dwelling size, and residential transitions move together (Mulder and Billari, 2010; Ström, 2010; Kulu and Steele, 2013; Chudnovskaya, 2019; Tocchioni et al., 2021). Demographic accounts of low fertility also emphasize broader changes in family formation, marriage, and living arrangements (Myrskylä et al., 2013; Lesthaeghe, 2014). I use that evidence as motivation for housing as a family-formation input, while the model below treats fertility as an economic choice over the lifecycle.

consumption, how much housing to retain, and the estate she leaves at exit. An old renter carries no housing and no capital-gains wedge.

Masses and demography. At the start of period t, a mass M_t of potential young households is drawn from a distribution $G_{Y,t}$. A potential young household either enters the modeled housing market or takes an outside option. Nonentrants live outside the modeled market in both periods. Current young entrants become next period's old households, and the next potential-young cohort is formed by births plus a renewal inflow R_t. The stationary cohort mass is denoted by $\bar{M}$.

Preferences. Children raise both consumption needs and housing needs. In the compact model fertility is continuous. A young household choosing consumption c_i, housing h_i, and fertility n_i has utility

$$u_i^Y(c_i, h_i, n_i) = \log(c_i - \chi n_i) + \alpha \log(h_i - \kappa n_i) + \vartheta \log n_i, \qquad \alpha, \vartheta, \chi, \kappa > 0. \tag{1}$$

The term χn_i is the goods requirement of children and κn_i is their housing-space requirement. A marginal child therefore uses the same housing stock as the parents.

Old households value consumption, housing, and bequests:

$$u^O(c_j^O, h_j^O, e_j) = \log c_j^O + \gamma \log h_j^O + \mathcal{B}(e_j), \qquad \gamma > 0, \tag{2}$$

where $e_j \geq 0$ is the estate and $\mathcal{B}$ is increasing and concave, possibly identically zero. Bequest motives and true old housing utility are preferences; they are not wedges.

Housing. There is a fixed stock $\bar{H}$ of divisible floorspace. Households occupy floorspace by renting or owning. The two tenure modes have different size limits:

$$\mathcal{H}^R = \{h : 0 < h \leq \bar{h}^R\}, \qquad \mathcal{H}^O = \{h : 0 < h \leq \bar{h}^O\}, \qquad 0 < \bar{h}^R < \bar{h}^O. \tag{3}$$

This is the family-housing segmentation: small units are available to rent, but large family-capable units are mainly reached through ownership. All floorspace trades in one market. The user cost of housing is q and the asset price is P, linked by

$$q = \rho P, \qquad \rho = r + \delta + \tau^p, \tag{4}$$

where r is the interest rate, δ is depreciation, and τ^p is the property-tax rate. For a given user cost q, a higher property tax lowers the asset price P and therefore lowers the down payment needed to buy a given house.

Financing. A buyer can finance at most share $\phi \in (0, 1)$ of the purchase price. Buying h units therefore requires the cash down payment

$$(1 - \phi)Ph. \tag{5}$$

Renters cannot borrow. Households can save in a bond at return $1 + r$. The compact model treats the down-payment constraint as an access constraint: it limits the amount of owner housing a young household can occupy.

Capital-gains taxation and step-up basis. Selling a house triggers a capital-gains tax. Dying with the house gives a stepped-up basis, so the accrued gain is forgiven. In a stationary real model, the gain comes from nominal price growth. If dollar prices grow at rate ϖ, a house bought one period ago has a taxable paper gain even when its real value is unchanged. The real flow-equivalent tax wedge per unit sold is

$$\bar{\ell} = \tau^g \frac{P\varpi}{1 + \varpi}, \tag{6}$$

after normalizing the period conversion factor to one. The wedge is not a preference and not a resource cost. It is a private tax liability avoided by holding the house until death.

4 Households

Conditional on entry, young household i chooses a tenure mode $m \in \{R, O\}$, housing, fertility, consumption, and savings. For $m \in \{R, O\}$,

$$W_i^m(q) = \max_{c_i, h_i, n_i, a_i'} \quad \log(c_i - \chi n_i) + \alpha \log(h_i - \kappa n_i) + \vartheta \log n_i \\ + \beta \left[\mathbf{1}\{m = R\} V^R(a_{j'}) + \mathbf{1}\{m = O\} V^O(a_{j'}, h_i) \right] \tag{7}$$

subject to

$$c_i + q h_i + a_i' = y_i + a_i, \tag{8}$$

$$h_i \in \mathcal{H}^m, \qquad a_i' \geq 0,$$

$$\mathbf{1}\{m = O\}(1 - \phi) P h_i \leq a_i, \tag{9}$$

$$c_i - \chi n_i > 0, \qquad h_i - \kappa n_i > 0.$$

Using $P = q/\rho$, the largest owner house feasible for type i is

$$H_i^O(q) = \min \left\{ \bar{h}^O, \frac{a_i \rho}{(1 - \phi) q} \right\}, \qquad H_i^R = \bar{h}^R. \tag{10}$$

H_i^m is the effective family-housing cap. A renter is capped by the rental stock. A buyer is capped by the owner size limit or by collateral, whichever is shorter.

An old incumbent enters with financial wealth a_j, carried housing H_j^0, and an accrued gain. She chooses consumption, retained housing, and an estate:

$$V^O(a_j, H_j^0) = \max_{c_j^O, h_j^O, e_j} \quad u^O(c_j^O, h_j^O, e_j) \\ \text{s.t.} \quad c_j^O + e_j + q h_j^O = a_j + q H_j^0 - \bar{\ell}(H_j^0 - h_j^O), \qquad 0 \leq h_j^O \leq H_j^0. \tag{11}$$

Selling one unit nets $q - \bar{\ell}$, so stepped-up basis acts as a subsidy to retaining the house. An old renter has no carried housing and no tax:

$$V^R(a_j) = \max_{c_j^O, h_j^O, e_j} u^O(c_j^O, h_j^O, e_j) \quad \text{s.t.} \quad c_j^O + e_j + q h_j^O = a_j, \qquad 0 < h_j^O \leq \bar{h}^R. \tag{12}$$

5 Entry and Equilibrium

A potential young household can enter the housing market or take an outside option with value $\bar{W}^E$. Entry has Type-I extreme-value shocks with scale κ_E. The inside value is

$$W_i^Y(q) = \max\{W_i^R(q), W_i^O(q)\},$$

and the entry probability is

$$\pi_i^E(q) = \frac{\exp\{W_i^Y(q)/\kappa_E\}}{\exp\{W_i^Y(q)/\kappa_E\} + \exp\{\bar{W}^E/\kappa_E\}}. \tag{13}$$

Let $n_i^Y(q)$ be fertility in the inside tenure chosen by type i, and let $\bar{n}_i^E$ be fertility in the outside option. Births generated by the potential-young cohort are

$$N(q) = \bar{M} \int \left[\pi_i^E(q) n_i^Y(q) + (1 - \pi_i^E(q)) \bar{n}_i^E \right] \mathrm{d}G_Y(i). \tag{14}$$

The baseline normalization sets $\bar{n}_i^E = 0$, but the entry-margin terms below are written so that outside fertility can be restored by replacing inside fertility with the gap $n_i^Y - \bar{n}_i^E$. In a stationary demographic equilibrium, births and renewal close the young cohort:

$$N + \bar{R} = \bar{M}. \tag{15}$$

The local formulas below hold $(\bar{M}, G_Y, F_O, F_R, \bar{R})$ fixed. A full demographic counterfactual instead recomputes the stationary fixed point, or else states which object is held fixed and which object absorbs the birth change.

Let F_O and F_R be the stationary measures of old incumbents and old renters. Aggregate floorspace demand is

$$H^D(q) = \bar{M} \int \pi_i^E(q) h_i(q) \, \mathrm{d}G_Y(i) + \int h_j^O(q) \, \mathrm{d}F_O(j) + \int h_j^O(q) \, \mathrm{d}F_R(j). \tag{16}$$

The compact equilibrium clears the single stock:

$$H^D(q) = \bar{H}, \qquad P = q/\rho. \tag{17}$$

Stationarity requires current young owners to become next period's old incumbents, current young renters to become old renters, and births plus the renewal inflow to reproduce the next young cohort. The transition rules assign old estates to entry wealth and are otherwise left general; the local results below condition on the stationary cross-section.

6 Equilibrium Wedges

The model's sufficient statistics are marginal values of space in consumption goods. Let λ_i^m be the multiplier on the young household's budget, θ_i^m the multiplier on the size limit, and μ_i the multiplier on the owner's down-payment constraint. For renters,

$$\frac{\alpha(c_i^R - \chi n_i^R)}{h_i^R - \kappa n_i^R} = q + \zeta_i^R, \qquad \zeta_i^R = \frac{\theta_i^R}{\lambda_i^R} \geq 0. \tag{18}$$

A capped renter values one more unit of housing above the rent it would command.

For young owners, anticipation of the future capital-gains liability raises the effective price of owning family-sized space. Under interior saving and interior old retention,

$$q^O = q + \frac{\bar{\ell}}{1+r}. \tag{19}$$

The owner access gap is

$$\zeta_i^O = \underbrace{\frac{\mu_i}{\lambda_i^O}(1-\phi)P}_{\zeta_i^{O,F}\text{, down-payment component}} + \underbrace{\frac{\theta_i^O}{\lambda_i^O}}_{\text{owner-size component}} \geq 0, \tag{20}$$

so the owner's marginal value of space is

$$\frac{\alpha(c_i^O - \chi n_i^O)}{h_i^O - \kappa n_i^O} = q^O + \zeta_i^O. \tag{21}$$

When the owner size limit is slack, $\zeta_i^O = \zeta_i^{O,F}$ and the gap is a financing gap.

The old incumbent's interior retention condition is

$$\frac{\gamma c_j^O}{h_j^O} = q - \bar{\ell}. \tag{22}$$

She keeps marginal space she values below the market user cost because selling it realizes the taxable gain. An old renter has no such wedge and, at an interior choice, values space at q.

The fertility condition prices the marginal child in goods. In tenure mode m, with $q^R = q$ and q^O given by (19),

$$\underbrace{\frac{\vartheta(c_i^m - \chi n_i^m)}{n_i^m}}_{\text{willingness to pay for a marginal child}} = \underbrace{\chi + \kappa(q^m + \zeta_i^m)}_{\text{full price of a child}}. \tag{23}$$

A child's space requirement is priced at the household's own shadow value of space. A binding access constraint therefore acts as an implicit child tax $\kappa\zeta_i^m$.

7 Efficiency and Fertility

The planner comparison holds the stationary cross-section, entry probabilities, tenure assignments, savings, and real size-limit constraints fixed at the instant of the variation. The planner counts true household preferences and true resource costs. It does not count the avoided capital-gains tax as a social benefit.

Proposition 1 (Local young-old housing misallocation)**.** *Consider a stationary equilibrium. Hold entry, tenure assignments, savings, and real size-limit constraints fixed. Take a young owner whose owner size limit is slack and whose financing constraint binds, so $\zeta_i^{O,F} > 0$, and an old incumbent at an interior retention choice. If one unit of existing housing can be reassigned from the old incumbent to the young owner at zero real implementation cost, the planner's goods-equivalent surplus is*

$$\underbrace{(q^O + \zeta_i^{O,F})}_{\textit{young buyer's marginal value}} - \underbrace{(q - \bar{\ell})}_{\textit{old incumbent's marginal value}} = \zeta_i^{O,F} + \frac{\bar{\ell}}{1+r} + \bar{\ell}. \tag{24}$$

If this expression is positive, the competitive allocation does not solve the planner's conditional allocation problem.

The market price cancels. What remains are the young household's financing gap, the buyer's anticipated gains tax, and the old incumbent's realized gains tax. The two tax terms are transfers, not resource costs. If implementing the reassignment requires real cost r_i^F per unit, that cost subtracts from the surplus. If the down-payment constraint is a primitive technological enforcement constraint and no instrument can relax it, the variation is not feasible.

Old occupancy is not inefficient by itself. An old owner who keeps a large house because she values it, or because she values leaving bequests, is using housing for reasons the planner also counts. The inefficiency is the part of retention supported by the forgiveness of capital-gains taxes at death, paired with young households who value family-sized space above its market user cost.

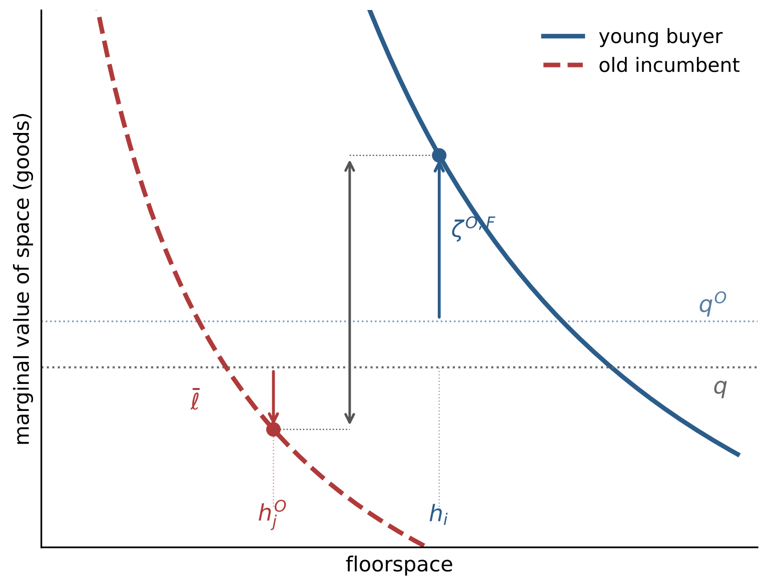

Figure 1: Local misallocation. A collateral-capped young buyer values marginal space at $q^O + \zeta^{O,F}$, while a locked-in old incumbent values retained space at $q - \bar{\ell}$. The vertical distance is the goods-equivalent surplus from a compensated marginal reassignment.

The same cap determines a fertility response. Fix a tenure mode and a household whose housing is pinned at the effective family-housing cap H. Concentrating the savings rule into the budget, fertility along the binding cap solves

$$\frac{\vartheta}{n} = \frac{(1+k)\chi}{w - q^m H - \chi n} + \frac{\alpha\kappa}{H - \kappa n},$$

where $w = y + a$ and k is the savings load on adult consumption. Differentiating with respect to the cap gives

$$\frac{\mathrm{d}n}{\mathrm{d}H} = \frac{\dfrac{\alpha\kappa}{(H-\kappa n)^2} - \dfrac{(1+k)\chi q^m}{(w - q^m H - \chi n)^2}}{\dfrac{\vartheta}{n^2} + \dfrac{(1+k)\chi^2}{(w - q^m H - \chi n)^2} + \dfrac{\alpha\kappa^2}{(H-\kappa n)^2}}. \tag{25}$$

Relaxing access raises fertility when the space-relief force dominates the budget-tightening force. A useful sufficient condition along every binding cap is

$$\chi \leq \frac{(1+k)\kappa q^m}{\alpha}. \tag{26}$$

The derivative connects the misallocation wedge to fertility: policies that raise the effective family-housing cap matter only for households whose binding constraint they actually relax.

This household derivative aggregates into a local sufficient statistic for births. Let z be a policy parameter. For renters, $H_i^R = \bar{h}^R$; for owners,

$$H_i^O(q) = \min\left\{\bar{h}^O, \frac{a_i \rho}{(1-\phi)q}\right\}.$$

Let $\mathcal{E}_m(z)$ be the set of types for whom tenure mode m is the unique active tenure, the effective cap H_i^m is strictly binding, and the binding component of the cap is locally unique. The pass-through of policy z to the effective family-housing cap is

$$\mathcal{P}_i^m(z) = \frac{\partial H_i^m(z)}{\partial z}.$$

For an exposed type, write residual adult consumption and residual space as

$$x_i^m = c_i^m - \chi n_i^m, \qquad s_i^m = h_i^m - \kappa n_i^m.$$

The fixed-mode fertility response to one more unit of permitted family housing is

$$\Psi_i^m = \frac{\dfrac{\alpha\kappa}{(s_i^m)^2} - \dfrac{\chi q^m}{(1+k)(x_i^m)^2}}{\dfrac{\vartheta}{(n_i^m)^2} + \dfrac{\chi^2}{(1+k)(x_i^m)^2} + \dfrac{\alpha\kappa^2}{(s_i^m)^2}}. \tag{27}$$

The value of relaxing the cap in utility units is

$$\Theta_i^m \equiv \frac{\partial W_i^m}{\partial H_i^m} = \frac{\alpha}{s_i^m} - \frac{q^m}{x_i^m} = \frac{\zeta_i^m}{x_i^m}. \tag{28}$$

Thus the same access gap that measures misallocation also governs the entry response.

Proposition 2 (Local fixed-state fertility statistic)**.** *Holding prices, tenure choices, active sets, outside fertility, and the stationary demographic state fixed, the local effect of a cap-shifting policy on cohort births is*

$$\left.\frac{\mathrm{d}N}{\mathrm{d}z}\right|_{q,\text{tenure},\mathcal{A}} = \bar{M} \sum_{m\in\{R,O\}} \int_{\mathcal{E}_m(z)} \left[\pi_i^E \Psi_i^m + (n_i^m - \bar{n}_i^E)\frac{\pi_i^E(1-\pi_i^E)}{\kappa_E}\Theta_i^m\right] \mathcal{P}_i^m(z)\,\mathrm{d}G_Y(i). \tag{29}$$

The first term is the intensive margin: already-entered households change completed fertility by Ψ_i^m for each unit of cap relief. The second term is the entry margin: cap relief raises the value of entering, and marginal entrants bring their inside fertility net of the fertility they would have had outside the modeled market. Both terms are gated by the same exposure set. A policy aimed at a slack constraint has zero pass-through in this channel.

The formula is fixed-tenure accounting. If tenure choice is deterministic and the type distribution has density near the rent-own boundary, a policy can also move households across that boundary. Let

$$\Delta_i(z) = W_i^O(z) - W_i^R(z), \qquad \mathcal{I}(z) = \{i : \Delta_i(z) = 0\},$$

and suppose $\nabla_i \Delta_i(z) \neq 0$ on $\mathcal{I}(z)$, with unique mode optima and no mass points at kinks. The tenure-switching component is

$$\left.\frac{\mathrm{d}N^{\text{switch}}}{\mathrm{d}z}\right|_{q,\mathcal{A}} = \bar{M} \int_{\mathcal{I}(z)} \pi_i^E \left(n_i^O - n_i^R\right) \frac{\partial_z \Delta_i(z)}{\|\nabla_i \Delta_i(z)\|} f(i)\,\mathrm{d}S(i), \tag{30}$$

where $f(i)$ is the type density and $\mathrm{d}S(i)$ is surface measure on the indifference set, oriented toward ownership. The total fixed-price effect is

$$\left.\frac{\mathrm{d}N}{\mathrm{d}z}\right|_{q} = \left.\frac{\mathrm{d}N}{\mathrm{d}z}\right|_{q,\text{tenure},\mathcal{A}} + \left.\frac{\mathrm{d}N^{\text{switch}}}{\mathrm{d}z}\right|_{q,\mathcal{A}}. \tag{31}$$

There is no separate entry-boundary term under logit entry because the inside value is continuous at the tenure boundary. The discontinuity is in conditional fertility, $\pi_i^E(n_i^O - n_i^R)$. If $q^O = q^R$, the two tenure modes differ only through caps and the boundary fertility jump disappears when the boundary allocation is the same in both modes. With a tax wedge, ownership is the more expensive way to access family space; the switching term is first order, but its sign depends on the fertility jump of boundary households.

Part II

Quantitative Lifecycle Model

8 Purpose and Timing

The quantitative model embeds the same mechanism in a finite-life economy. It has the standard lifecycle structure: households age, face income risk, save, choose whether to rent or own, choose housing size, decide fertility, and eventually leave bequests. The housing market has one aggregate user cost and one asset price. Tenure matters because it changes the size menu, financing constraints, transaction costs, and tax exposure.

Time is discrete. Households enter adult life at age $a = 1$, retire after age J_R, and die at the end of age J. The individual state at the beginning of a period is

$$x = (b, z, d, a, n, s). \tag{32}$$

Here b is liquid wealth, $z \in \mathcal{Z}$ is idiosyncratic productivity, d is tenure and housing position, a is age, n is completed fertility, and s is the child-at-home state. Tenure belongs to

$$\mathcal{D} = \{R, O_1, \dots, O_K\}. \tag{33}$$

R denotes renting. O_k denotes ownership of a house with physical size H_k, where $H_1 < \cdots < H_K$. Renters choose housing continuously up to $\bar{h}^R$; owners choose from the discrete ladder. The ladder is the quantitative version of the family-housing segmentation in Part I.

Income for working households is

$$y(a, z) = (1 - \tau_{\text{pay}}) w e_a z, \qquad z' \sim \Pi^z(\cdot|z), \qquad a \le J_R, \tag{34}$$

and retired households receive pension income p^{ret}. The wage is common apart from age and productivity because the quantitative mechanism is housing access, tenure, and lifecycle wealth.

9 Preferences and Bequests

Let $m(n, s)$ be the number of dependent children living at home. Completed fertility n and current dependents differ because children eventually mature. Household needs are

$$\begin{aligned} \bar{c}(n, s) &= \bar{c}_0 + \bar{c}_n m(n, s), \\ \bar{h}(n, s) &= \bar{h}_0 + \bar{h}_{\text{jump}} \mathbf{1}\{m(n, s) \ge 1\} + \bar{h}_n m(n, s). \end{aligned} \tag{35}$$

The first child can create a discrete housing-space jump; additional dependent children raise the required space linearly.

Flow utility is Stone-Geary over consumption and housing services, wrapped in CRRA:

$$u(c, \mathbf{h}; n, s) = \frac{\left[(c - \bar{c}(n, s))^{\alpha}(\mathbf{h} - \bar{h}(n, s))^{1-\alpha}\right]^{1-\sigma}}{1 - \sigma} + \psi_{\text{child}} m(n, s), \tag{36}$$

with utility equal to $-\infty$ when either residual is nonpositive. Housing services differ by tenure:

$$\mathbf{h} = \begin{cases} h^R, & d = R, \\ \chi_O H_k, & d = O_k, \end{cases} \qquad \chi_O > 0. \tag{37}$$

In market clearing, owners demand the physical quantity H_k, not the service quantity $\chi_O H_k$.

At death, households receive warm-glow bequest utility. Let W^B be bequeathable wealth after liquidation costs and taxes. Bequest utility is

$$\mathcal{B}(W^B, n) = \theta_0(1 + \theta_n n)\frac{(\theta_1 + \max\{W^B, 0\})^{1-\sigma} - \theta_1^{1-\sigma}}{1 - \sigma}. \tag{38}$$

The parameter θ_n lets completed fertility affect the desired estate even after children have left the household.

10 Housing, Finance, and Taxes

There is one aggregate housing-services market. The user cost q and asset price P satisfy

$$q = (r + \delta + \tau^p)P. \tag{39}$$

Housing supply is upward sloping:

$$H^S(q) = H_0\left(\frac{q}{\bar{q}}\right)^{\eta}. \tag{40}$$

The fixed-stock compact model is the limiting case $\eta = 0$ around the benchmark stock.

Renters choose c, h^R, b' subject to

$$c + qh^R + b' = Rb + y(a, z), \qquad 0 \le h^R \le \bar{h}^R, \qquad b' \ge 0, \tag{41}$$

where $R = 1 + r$. Owners of rung H_k choose c, b' subject to the owner branch budget and the borrowing limit described below. Owners pay maintenance and property taxes through $(\delta + \tau^p)PH_k$.

At origination, a buyer can finance at most share ϕ of house value, so the required down payment is

$$(1 - \phi)PH_k. \tag{42}$$

The post-purchase liquid position must satisfy

$$b' \ge -\phi PH_k. \tag{43}$$

An incumbent owner who keeps the same house is not forced to delever after a price change:

$$b' \ge \min\{b, -\phi PH_k\}. \tag{44}$$

This convention constrains origination but lets incumbency carry state dependence.

Housing adjustment is costly. Let ψ^s be a proportional sale cost. When the capital-gains block is active, the state of an owner also includes a tax basis B^p. A sale of house H_k triggers tax

$$\mathcal{T}^g(PH_k, B^p) = \tau^g \max\{PH_k - B^p - \bar{g}, 0\}, \tag{45}$$

where $\bar{g}$ is the statutory exclusion. Net sale proceeds are

$$\Omega(H_k, B^p) = (1-\psi^s)PH_k - \mathcal{T}^g(PH_k, B^p). \tag{46}$$

At death, stepped-up basis resets the heir's basis to market value, so accrued gains are not taxed. This is the quantitative counterpart of the old-owner lock-in wedge $\bar{\ell}$ in Part I.

11 Fertility and Child Aging

Fertility is chosen during the fertile window

$$a \in \{A_f^{\text{start}}, \dots, A_f^{\text{end}}\}.$$

A childless household chooses

$$n' \in \mathcal{N} = \{0, 1, \dots, \bar{n}\}. \tag{47}$$

The option $n' = 0$ means no birth in the current period. A positive choice fixes completed fertility forever and moves the household into the first dependent-child state. A household that reaches the end of the fertile window without a positive choice remains childless.

Let $s = 0$ denote no dependent children and let $s = 1, \dots, S_c$ denote dependent-child stages. After a positive fertility choice, the household enters $s = 1$. Children age stochastically through the stages according to Π^c. While children are at home, $m(n, s) = n$; after maturity, $m(n, s) = 0$. Completed fertility still matters through bequest utility.

12 Household Problem

Within the period, the household first faces fertility if eligible, then chooses tenure and housing position, then chooses consumption, rental housing, and saving. Let the continuation value after income risk and child aging be

$$\bar{V}_{a+1}(b', d', z, n, s) = \sum_{z'} \sum_{s'} \Pi^z(z'|z)\Pi^c(s'|s, n)V_{a+1}(b', z', d', n, s'). \tag{48}$$

The terminal value is

$$V_{J+1}(b, z, d, n, s) = \mathcal{B}(W^B, n).$$

Conditional on renting, the branch value is

$$\begin{aligned} V_a^R(b, z, n, s) = \max_{c, h^R, b'} \quad & u(c, h^R; n, s) + \beta \bar{V}_{a+1}(b', R, z, n, s) \\ \text{s.t.} \quad & c + qh^R + b' = Rb + y(a, z), \\ & 0 \le h^R \le \bar{h}^R, \qquad b' \ge 0. \end{aligned} \tag{49}$$

Conditional on owning rung H_k after any adjustment, the branch value is

$$\begin{aligned} V_a^{O_k}(\tilde{b}, z, n, s) = \max_{c,b'} \quad & u(c, \chi_O H_k; n, s) + \beta \bar{V}_{a+1}(b', O_k, z, n, s) \\ \text{s.t.} \quad & c + (\delta + \tau^p) P H_k + b' = R\tilde{b} + y(a, z), \\ & b' \geq \underline{b}_k^O. \end{aligned} \tag{50}$$

The lower bound $\underline{b}_k^O$ is the origination or incumbent constraint in (43)–(44), depending on whether the household buys or keeps the current house.

Let $H(R) = 0$ and $H(O_k) = H_k$. The liquid wealth available after housing adjustment is

$$\tilde{b}(b, d, d') = b + \mathbf{1}\{d \in \mathcal{O},\ d' \neq d\}\Omega(H(d), B^p) - \mathbf{1}\{d' \in \mathcal{O},\ d' \neq d\}PH(d'). \tag{51}$$

Without the capital-gains block, $\Omega(H(d), B^p) = (1 - \psi^s)PH(d)$.

The tenure branch value for feasible d' is

$$\widetilde{V}_a^T(d'|b, z, d, n, s) = \begin{cases} V_a^R(\tilde{b}(b, d, R), z, n, s), & d' = R, \\ V_a^{O_k}(\tilde{b}(b, d, O_k), z, n, s), & d' = O_k. \end{cases} \tag{52}$$

Tenure and owner-rung choices have Type-I extreme-value shocks with scale κ_T:

$$\pi_a^T(d'|x) = \frac{\exp\{\widetilde{V}_a^T(d'|x)/\kappa_T\}}{\sum_{\hat{d} \in \mathcal{D}(x)} \exp\{\widetilde{V}_a^T(\hat{d}|x)/\kappa_T\}}, \tag{53}$$

and the inclusive tenure value is

$$V_a^T(x) = \kappa_T \log \sum_{\hat{d} \in \mathcal{D}(x)} \exp\{\widetilde{V}_a^T(\hat{d}|x)/\kappa_T\}. \tag{54}$$

For a childless household in the fertile window, the value of fertility option n' is

$$\widetilde{V}_a^F(n'|b, z, d) = V_a^T(b, z, d, n', s(n')), \qquad s(n') = \mathbf{1}\{n' > 0\}. \tag{55}$$

With Type-I extreme-value shocks of scale κ_F,

$$\pi_a^F(n'|b, z, d) = \frac{\exp\{\widetilde{V}_a^F(n'|b, z, d)/\kappa_F\}}{\sum_{\hat{n} \in \mathcal{N}} \exp\{\widetilde{V}_a^F(\hat{n}|b, z, d)/\kappa_F\}}. \tag{56}$$

Outside the fertile window, or after a positive fertility choice, this stage is degenerate and $V_a = V_a^T$.

13 Stationary Equilibrium

Let g be the stationary distribution of adult households over states. Adult mass is normalized to one for the benchmark quantitative exercise:

$$\int dg(x) = 1. \tag{57}$$

This normalization fixes the demographic envelope and lets fertility change household composition, tenure, and housing demand. It is not, by itself, a full demographic closure for total births. The quantitative counterfactuals therefore distinguish per-capita completed-fertility effects from any local birth or market-scale effect that operates through entry.

Definition 1 (Stationary equilibrium)**.** *A stationary equilibrium is a user cost and asset price (q, P), value functions and policies, tenure and fertility probabilities, a stationary distribution g, and aggregate housing quantities (H^D, H^S) such that:*

1. *households solve the branch problems (49)–(50), tenure choice is given by (53), fertility choice is given by (56), and terminal utility is (38);*
2. *the distribution g is invariant under policies, income transitions, fertility shocks, child aging, deterministic aging, death, and entry of new adult households;*
3. *aggregate physical housing demand is*

$$H^D = \int \left[\mathbf{1}\{d = R\} h^R(x) + \sum_{k=1}^{K} \mathbf{1}\{d = O_k\} H_k \right] \mathrm{d}g(x); \tag{58}$$

4. *the housing market clears and the user-cost relation holds:*

$$H^D = H^S(q), \qquad q = (r + \delta + \tau^p)P; \tag{59}$$

5. *payroll taxes, property taxes, capital-gains taxes, pensions, and any rebates satisfy the government accounting rule specified for the calibration or counterfactual.*

14 Calibration Objects

The quantitative model should be disciplined by the margins that identify the housing-fertility mechanism. The table gives the paper-facing organization of the calibration; it is not a results table.

Table 1: Parameter blocks and target moments

Block	Parameters	Main moments
Preferences and fertility	$\beta, \sigma, \alpha, \psi_{\text{child}}, \kappa_F$	completed fertility, childlessness, timing of first birth, wealth profiles
Child costs	$\bar{c}_n, \bar{h}_{\text{jump}}, \bar{h}_n$	rooms or housing services around first birth and additional births
Tenure and housing services	$\chi_O, \bar{h}^R, \{H_k\}, \kappa_T$	ownership by age, parent ownership gap, renter-owner size gap
Credit and adjustment	ϕ, ψ^s	ownership transition, liquid-wealth constraints, sale and downsizing rates
Bequests	$\theta_0, \theta_n, \theta_1$	old-age ownership, wealth at death, parent-childless saving differences
Capital gains and lock-in	$\tau^g, \bar{g}, B^p$	old-owner retention, downsizing around gains exposure, tax basis moments
Housing supply	$H_0, \eta, \bar{q}$	housing expenditure share, aggregate housing services, price response

The moment table should separate model moments from calibration targets. The natural blocks are fertility, lifecycle ownership, housing-size responses to children, old-owner retention, wealth and bequests, and market-clearing quantities.

15 Counterfactuals

The policy experiments should map directly into the primitive access margins:

1. relax down-payment constraints by increasing ϕ or giving targeted down-payment resources to young families;
2. expand family-sized rental access by increasing $\bar{h}^R$ or reducing the premium on large rental units;
3. reduce transaction costs ψ^s;
4. reform capital-gains taxation and step-up basis by changing τ^g, $\bar{g}$, or the treatment of gains at death;
5. increase housing supply through H_0 or η;
6. combine supply expansion with access policies to separate price effects from allocation effects.

Each counterfactual should report fertility, childlessness, ownership, young-family housing, old-owner retention, prices, and welfare. The compact model says where the effects should be strongest: households with positive family-housing access gaps and policies that pass through to the binding cap.

16 What Carries Over from the Compact Model

The quantitative model changes the state space but not the mechanism. In the compact model, a constrained young household prices a child's housing need at $q^m + \zeta_i^m$, while an old locked-in owner values retained marginal space at $q - \bar{\ell}$. In the quantitative model these are not single closed-form objects for every state, but they remain the diagnostic objects:

$$\text{young family-space shadow value} \quad \text{and} \quad \text{old retention shadow value.}$$

The calibration must therefore discipline not only aggregate fertility and aggregate ownership, but also the distribution of housing space by age, tenure, and child status. A model that matches total ownership while putting too much family-sized housing with unconstrained households would miss the mechanism.

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